Resolution of the FinalCorruptZip Forensic Challenge

Author: David Liu, Bennett Plumb, Devkumar Banerjee

Challenge Overview

The FinalCorruptZip challenge involves repairing a corrupted ZIP archive and a modified PNG file to recover a hidden flag (SVBGR{m4g1C_B1t3s_yUmmmmm}). The solution requires correcting file signatures (magic bytes) at both the ZIP and PNG levels. Below is a detailed breakdown of the resolution process.

Tools and Prerequisites

Hex Editors:
HxD (Windows), xxd/hexedit (Linux), or Bless (macOS/Linux) for binary file analysis.
Critical for direct byte-level modifications.
Archive Utilities:
7-Zip, WinZip, or command-line tools like unzip (Linux/macOS) for extraction.
Image Validation Tools:
Built-in OS image viewers or tools like IrfanView to confirm PNG integrity.

Step 1: Repairing the ZIP File Header

Background

ZIP files are identified by their magic bytes 50 4B 03 04 (ASCII: PK..). In this challenge, the first byte (50, representing "P") was altered to 41 ("A"), rendering the file unreadable by standard extractors.

Procedure

Initial Analysis:
Open corrupted.zip in a hex editor.
Observe the first four bytes: 41 4B 03 04 (incorrect) instead of 50 4B 03 04 (correct).

Byte Correction:
Navigate to offset 0x00 (first byte).
Replace 41 with 50 to restore the ZIP signature.
Save the modified file as repaired.zip.
Validation:
Attempt extraction using unzip repaired.zip (Linux/macOS) or 7-Zip (Windows).
Successful extraction yields a file named image.png. If extraction fails, re-verify the hex values or check for additional corruption.

Step 2: Repairing the PNG Signature

Background

PNG files begin with an 8-byte signature: 89 50 4E 47 0D 0A 1A 0A. The first four bytes in this challenge were altered to 65 4C 4F 4C (ASCII: eLOL), which must be corrected to 65 50 4E 47 (ePNG) to restore readability.

Procedure

Hex Analysis of image.png:
Open the extracted image.png in a hex editor.
Search for the sequence 65 4C 4F 4C at offset 0x00.
Signature Correction:
Replace the first four bytes as follows:
Original (corrupted): 65 4C 4F 4C
Corrected: 65 50 4E 47
Save the modified file as repaired_image.png.
Validation:
Open repaired_image.png in an image viewer. If the flag is not visible:
Confirm the full 8-byte PNG header: 89 50 4E 47 0D 0A 1A 0A.
Use file repaired_image.png (Linux/macOS) to verify the file type.

Flag Retrieval

Upon successful repair of both files, open repaired_image.png to view the flag:
SVBGR{m4g1C_B1t3s_yUmmmmm}

Technical Insights and Lessons Learned

Magic Bytes Significance:
File parsers rely on magic bytes to identify formats. Even minor alterations (e.g., changing P to A) disrupt parsing.
Hex Editing Best Practices:
Always create backups before modifying files.

Use checksums (e.g., sha256sum) to detect unintended changes.

Systematic Validation:

Test fixes incrementally. For example, repair the ZIP first, confirm extraction, then address the PNG.

Recommendations for Further Skill Development

Practice file forensics on platforms like CyberChef or CTF sites (e.g., CTFtime).

Study common file signatures (e.g., PDF: 25 50 44 46, JPEG: FF D8 FF E0).

Conclusion

This challenge underscores the importance of understanding low-level file structures and the role of magic bytes in digital forensics. By methodically correcting corrupted headers, analysts can recover data even from severely damaged files.